# Digital Audio-Video Signal Processing

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## 1 Question - 8 pts

Describe the Lukas-Kanade optical flow estimation algorithm. Explain under which circumstances the optical-flow estimation problem is ill-posed and how this situation can be detected in practice.

Consider the following blocks of pixels

| | | |
|---|---|---|
| 3 | 2 | 1 |
| 2 | 2 | 1 |
| 3 | 2 | 1 |

Frame $t$

| | | |
|---|---|---|
| 4 | 3 | 1 |
| 4 | 4 | 2 |
| 4 | 3 | 2 |

Frame $t-1$

Compute the optical flow estimate for the central pixel. You can approximate first order derivatives using discrete operators (e.g. the horizontal gradient is: $I_x(x,y) = I(x,y) - I(x-1,y)$). Check if a reliable estimate can be provided for both the horizontal and vertical displacement.

## 2 Question - 6 pts

Introduce the KLT transform, highlighting its properties and limitations in practical application scenarios.

## 3 Question - 6 pts

Illustrate the block diagram of an analysis-by-synthesis speech codec. List all the parameters that needs to be transmitted to the decoder and how they are quantized.

# 4 Question - 12 pts

Let $\mathbf{x}$ denote a vector of $n = 10000$ samples of a Gaussian distribution ($\mu_x = 0$, $\sigma_x = 1$).

- Implement a uniform scalar quantizer and quantize the data samples $\mathbf{x}$ using $R = 1, 2, \ldots, 8$ bits/sample.
- Plot the empirical rate distortion function (SNR vs. R) obtained quantizing $\mathbf{x}$.
- Compare the empirical rate-distortion function with Shannon's bound for Gaussian sources.
- Carefully adjust the size of the granular region to obtain a good coding efficiency over the whole range of rates.
- Implement a compander-expander quantizer. You might find the Matlab functions `normcdf` and `norminv` useful.
- Adjust the parameter $SIGMA$ of the `normcdf` and `norminv` functions in order to achieve the best rate distortion performance (hint: set $SIGMA = k * \sigma_x, k >= 1$).